

A Group Index Measurement Scheme using Mach-Zender Interferometer Arrays ¹

Md Nafiz Amin, University of California

Abstract—A waveguide group index measurement scheme based on an array of asymmetric Mach-Zehnder interferometers (MZIs) is proposed. This approach exploits the relationship between the experimentally measured free spectral range (FSR) of an MZI and the group index of the waveguide. By incorporating multiple MZIs with different path-length difference between adjacent arms, the scheme provides group index measurements over a 50 nm bandwidth in 1550-1600 nm wavelength range. The performance of the proposed scheme will be experimentally verified through fabrication and optical characterization of the MZI array, with the measured results compared against numerical predictions.

I. INTRODUCTION

MACH-Zender interferometers (MZIs) are fundamental photonic components that enable precise characterization of optical phase and propagation effects by comparing the optical fields traveling through two separate paths. Beyond their use in optical modulation and signal processing, MZIs can serve as effective characterization structures for extracting important waveguide properties. In particular, the spectral periodicity of an unbalanced MZI is directly related to the group index and a well-defined path-length difference between its two arms. This provides a convenient approach for experimentally determining the group index of a waveguide. Based on this, we propose a group index measurement scheme based on an array of asymmetric MZIs with different path-length differences.

II. MZI WORKING PRINCIPLE

An MZI consists of two Y-branch splitters connected by two waveguide arms. In a typical silicon-on-insulator (SOI) implementation, the input optical field is divided equally between the two arms and subsequently recombined by the second Y-branch splitter. When the two arms have different propagation lengths, the optical fields accumulate different phase shifts during propagation. The resulting phase difference leads to constructive or destructive interference at the output, producing a wavelength-dependent transmission response.

For an input electric field E_i , the fields propagating through the two arms can be expressed as

$$E_1 = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1}, E_2 = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2} \quad (1)$$

Here, L_1 and L_2 are the respective arm lengths and $\beta = \frac{2\pi}{\lambda} n_{eff}$ is the propagation constant. After recombination, the output field is

$$E_o = \frac{E_i}{2} (e^{-i\beta L_1} + e^{-i\beta L_2}). \quad (2)$$

This gives the normalized transfer function of an MZI as

$$I_o/I_i = \frac{1}{2} (1 + \cos(\beta \Delta L)). \quad (3)$$

where, $\Delta L = L_1 - L_2$ is the path-length difference. Thus, the transmission spectrum for the MZI shows wavelength-dependent resonance dips and peaks. As it will be derived later, free spectral range (FSR), is determined by ΔL and the waveguide group index n_g , which is used as the basis for the group index measurement scheme.

III. SIMULATION AND MODELING

B. Waveguide Simulation

The waveguide in the proposed scheme is a SOI (silicon-on-insulator) waveguide surrounded by silicon oxide layer on all sides as the cladding. The waveguide targeted film thickness is 220 nm and width is 500 nm. Fig. 1 shows the TE and TM mode intensity profiles, respectively, at 1550 nm. For the purpose of the MZI design, only TE mode is considered.

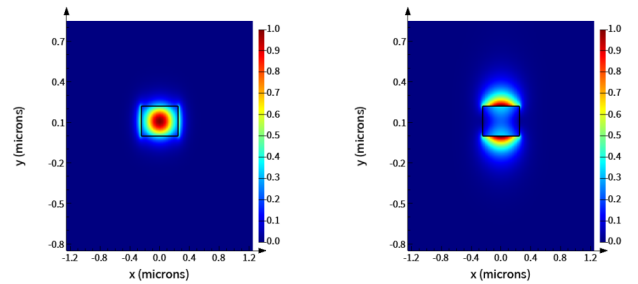


Fig. 1. Simulated fundamental TE (left) and TM (right) modes intensity profile.

Using Lumerical Modesolver, the effective index n_{eff} and group index n_g of the fundamental TE mode is also calculated in the wavelength range 1550 to 1600 nm. Figs. 2,3 show the change effective and group index, respectively, over the wavelength range. Using this data, a compact model for the waveguide model can be expressed as

$$n_{eff} = 2.444 - 1.13(\lambda - \lambda_o) - 0.0395(\lambda - \lambda_o)^2. \quad (4)$$

This compact waveguide model can be used in a circuit solver such as Lumerical Interconnect to simulate the MZI spectral characteristics.

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B. Group Index measurement form MZI

For an unbalanced Mach-Zehnder interferometer (AMZI) with a path-length difference ΔL , the phase difference between the two arms is

$$\Delta\varphi(\lambda) = \frac{2\pi}{\lambda} n_{eff} \Delta L \quad (5)$$

For an m-th order destructive interference, $\Delta\varphi(\lambda) = (2m + 1)\pi$. The phase difference between two adjacent dips separated in wavelength by FSR is

$$\Delta\varphi(\lambda + \text{FSR}) - \Delta\varphi(\lambda) = 2\pi \quad (6)$$

For a small FSR, it can be alternatively expressed as a derivative $\frac{d\Delta\varphi}{d\lambda} \text{FSR} \approx 2\pi$. Differentiating both sides of Eq. 5 gives

$$\frac{d\Delta\varphi}{d\lambda} = 2\pi \Delta L \frac{d}{d\lambda} \left(\frac{n_{eff}}{\lambda} \right) \quad (7)$$

Using the definition $n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$, Eq. (7) can be rewritten as

$$\left| \frac{d\Delta\varphi}{d\lambda} \right| = \frac{2\pi n_g \Delta L}{\lambda^2} \quad (8)$$

Substituting this into the derivative form of Eq. 6 gives

$$n_g = \frac{\lambda^2}{\text{FSR} \cdot \Delta L} \quad (9)$$

Thus, a measurement of one FSR can produce an initial value of n_g . This may be sufficient when $\frac{dn_g}{d\lambda}$ is negligible, i.e., n_g at the midpoint of the measurement bandwidth is a representative of the entire bandwidth. In practice, the whole transmission spectrum of an array of AMZIs with different ΔL , and each with more than one interference fringe in the spectrum, is used to measure n_g .

As an example, a particular AMZI with $N+1$ interference fringes at locations $\lambda_1, \lambda_2, \dots, \lambda_{N+1}$ within the measurement spectrum has N unique FSRs. A window with M fringes can be slid in one fringe steps across the spectrum to get FSR and n_g at the center wavelength λ_c of that window. For an x-th fringe in the spectrum, this can be expressed as

$$\text{FSR}_x^{(M)} = \frac{\lambda_{x+M} - \lambda_x}{M} \quad (10)$$

This provides an estimation of $n_{g,x}(\lambda)$ at the center wavelength of the M-fringe window $\lambda_{c,x} = (\lambda_{x+M} + \lambda_x)/2$. Thus, an $N-M+1$ point plot of n_g vs λ can be obtained. A small window may give better spectral resolution but prone to.

TABLE I

ESTIMATION OF SPECTRAL BEHAVIOR VS ΔL

Design	ΔL [μm]	#FSR $\approx N$ in measurement bandwidth	Simulated FSR [nm] @1575 nm	Estimated FSR [nm] w/ simulated n_g @1575 nm
AMZI 1	100	8.47	5.86	5.91
AMZI 2	150	12.7	3.89	3.93
AMZI 3	200	16.93	2.97	2.95
AMZI 4	250	21.17	2.36	2.36
AMZI 5	300	25.4	1.96	1.96
MZI 0	0	-	-	-

more measurement noise and vice versa, thus this trade-off will need to be considered.

III. GUIDELINES FOR MANUSCRIPT PREPARATION

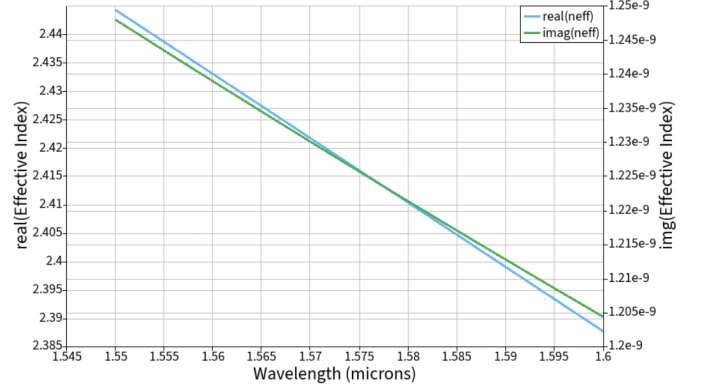


Fig. 2. Effective index vs wavelength for TE00 mode.

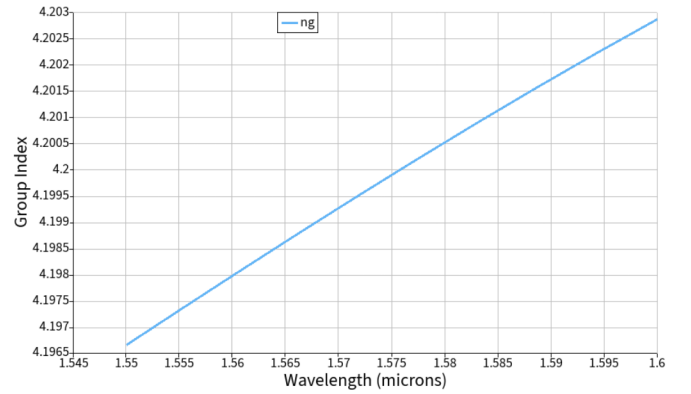


Fig. 3. Group index vs wavelength for TE00 mode.

The experimental measurement will be performed within the bandwidth 1.55 μm to 1.6 μm in 1 pm laser steps. Using the average simulated value of n_g in Eq. 9 and integrating over this bandwidth, the number of FSRs, N can be approximated as

$$N = \int_{1.55}^{1.6} \frac{n_g \Delta L}{\lambda^2} d\lambda \approx 0.084672 \Delta L [\mu\text{m}] \quad (9)$$

The values of N for five candidate AMZI designs with different proposed ΔL are listed in Table 1. To check the fabrication quality, a balanced MZI is also included in the plan, which should give a flat spectral response.

C. MZI Circuit simulation

The transmission spectra of the proposed MZI circuit for different values of ΔL are simulated using Lumerical Interconnect. High-volume experimental testing in a PIC is typically done using grating couplers, which are added for TE mode in these MZI simulations to include their spectral response (shown in Fig. 4). Furthermore, to approximate the actual waveguide loss and dispersion behavior, SiEPIC PDK was used in these simulations. Fig. 5 shows the gain curve of an MZI with $\Delta L = 100 \mu\text{m}$. As expected from the previous calculations, 9 resonance fringes are observed in the simulated bandwidth. The slow roll-off of the MZI gain curve is also

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observed when the grating coupler transmission spectrum is factored in.

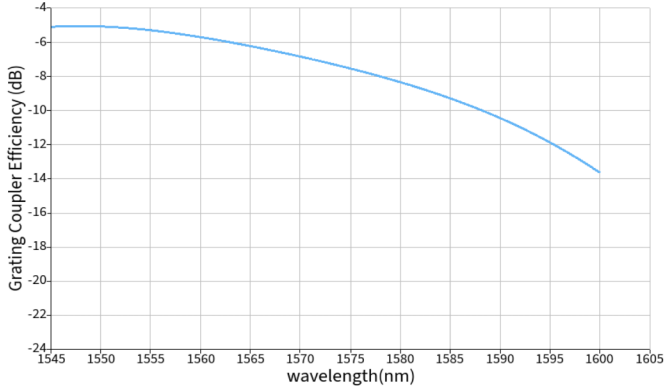


Fig. 4. Effective index vs wavelength.

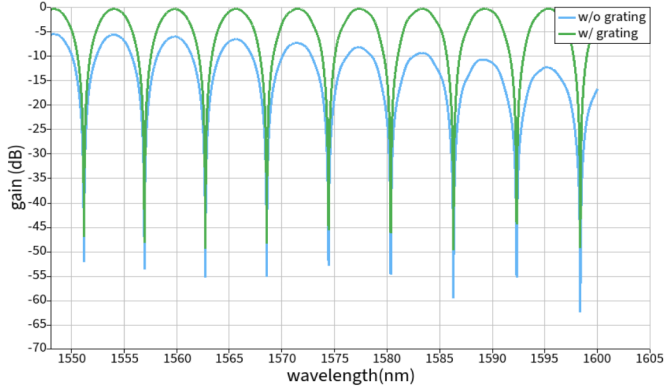


Fig. 5. Group index vs wavelength.

From these simulations, the estimated FSRs at the center wavelength of the measurement bandwidth is listed in Table 1. The values are in good agreement with the calculated FSR using the simulated group index n_g at the center wavelength from the mode solver. It's worth mentioning that the experimental laser step capability of 1 pm (~ 121 MHz) is sufficient to resolve the smallest FSR of 1.96 nm (~ 0.23 GHz), with ~ 1900 spectral sample points in between. Finally, the sinusoidal transmission spectra of the same MZI is shown in Fig. 6, demonstrating the behavior predicted by Eq. (3).

IV. SIMULATION AND MODELING

The proposed scheme shows good behavior agreement between mode solver and interconnect simulations, and the planned design variations is verified to be well within the fabrication and measurement capabilities. The experimental data will be validated against the simulation results in the final report.

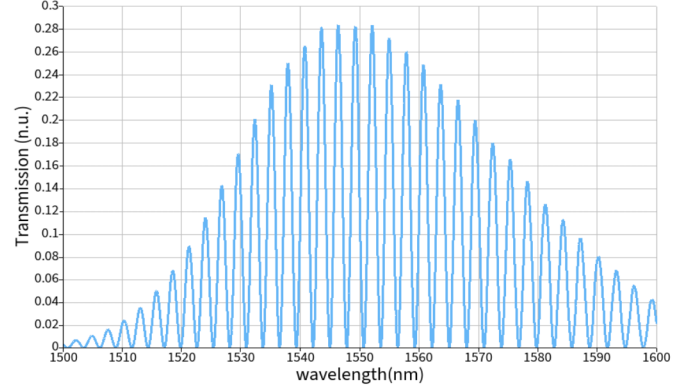


Fig. 6. MZI transmission spectrum with two grating couplers